

HARSH SHROFF

AI/ML ENGINEER

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PROFESSIONAL SUMMARY

AI/ML Engineer specializing in architecting production-grade RAG pipelines, **Neuro-Symbolic AI agents**, and multi-model orchestration for enterprise systems. Proven expertise in building verifiable, auditable AI workflows and deploying low-latency LLM inference engines across distributed cloud environments. Adept at designing scalable data pipelines, establishing strict ML governance, and leveraging advanced vector search techniques to solve complex data challenges. Strong track record of transitioning legacy manual processes into secure, automated reasoning engines that drive measurable business efficiency.

CORE COMPETENCIES

Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-Learn, Neural Networks, Computer Vision (YOLOv8, OpenCV), NLP, Time Series Analysis, Fine-Tuning, RLHF

GenAI & LLM Ecosystem: LLM Agents, Neuro-Symbolic AI, **PydanticAI**, LangChain, LlamaIndex, RAG, Prompt Engineering, Claude 3.5, Llama 3.1, Gemini 2.0, OpenAI API, Ollama, Hugging Face

Cloud & MLOps: AWS (Bedrock, SageMaker, EC2, S3, Textract, Lambda), MLflow, Weights & Biases, Docker, GitHub Actions (CI/CD), TensorRT, Model Governance, LLM Observability

Software & Infrastructure: Python, FastAPI, Linux/Bash, Git, SQL, PostgreSQL, Redis, Supabase, Vector Databases (Pinecone/pgvector), RESTful APIs, Asynchronous Processing (APScheduler)

Data Engineering: Pandas, NumPy, Apache Spark, Hadoop, Feature Engineering, ETL/ELT Pipelines, Data Modeling, Semantic Search

PROFESSIONAL EXPERIENCE

AI/ML Researcher

Mar 2023 – Present

UMBC Center for Real-time Distributed Sensing and Autonomy

- **Architected a capability-based multi-agent AI system** deploying quantized Llama 3.1 models via Ollama, achieving sub-8s end-to-end latency for autonomous reasoning and voice-controlled execution.
- **Invented a patent-pending multi-modal AI assistant** synthesizing real-time sensor data fusion (RGB, thermal, audio) under US Army Research Lab funding, improving real-time optimization systems by **20%**.
- **Engineered highly optimized machine learning workflows** leveraging TensorRT acceleration to achieve 50+ FPS object detection and zero-latency continuous scene understanding.
- **Developed an asynchronous threat-monitoring pipeline** featuring state-tracking, dynamic LLM inference, and automated priority overrides without manual intervention.

AI/ML Engineer - Production AI Systems

Jun 2024 – May 2025

VITG Corp., Halethorpe, MD

- **Architected a public-facing civic RAG platform** hosted on EC2, processing 2,500+ municipal regulatory documents to help citizens assess property compliance against dynamic geospatial lake boundaries.
- **Designed the semantic search architecture** for the civic platform using S3, vector databases, and LLMs to parse unstructured zoning data, ensuring high-accuracy retrieval for high-stakes demolition queries.
- **Engineered a serverless AI Slackbot** utilizing AWS Bedrock and Lambda to automate internal HR workflows, enabling recruiters to perform dynamic, LLM-driven candidate evaluation and reducing screening time by **60%**.
- **Established ML governance frameworks** and scalable ETL pipelines across both deployments, utilizing automated evaluation metrics to mitigate hallucinations and maintain production reliability. *FastAPI, PostgreSQL, Docker*

Data Scientist Intern - Computer Vision

Aug 2023 – Dec 2023

The Conservation Fund, Shepherdstown, WV

- **Deployed an end-to-end computer vision ML pipeline** on edge devices (Raspberry Pi), achieving **92% accuracy** in real-time quality assessment using YOLOv8 and OpenCV. *Computer Vision, Edge Deployment, MLOps*
- **Co-authored peer-reviewed research** demonstrating the commercial impact of deep learning applications, enabling the deployment of AI quality control systems with automated reporting capabilities.

WeHear Innovations Pvt. Ltd., Ahmedabad, India

- **Developed Personal Hearing Intelligence (PHI) models**, analyzing longitudinal audio exposure and ambient noise patterns to generate personalized hearing-risk metrics using machine learning.
- **Built time-series feature engineering pipelines** for high-frequency sensor data, enabling early-risk detection through statistical monitoring of volume and frequency band exposure.
- **Optimized audio signal processing workflows** for bone-conduction hardware, implementing low-latency ML inference to support real-time voice amplification features.

KEY TECHNICAL PROJECTS

Silicon Oracle: Asynchronous LLM Intelligence Engine ([GitHub](#))

Python, Flask, PostgreSQL, Gemini 2.0 Flash, Finnhub API, APScheduler

- Engineered a multi-tenant asynchronous ML pipeline utilizing APScheduler and PostgreSQL to ingest real-time market data and orchestrate parallelized LLM inference via Gemini 2.0 Flash.
- Designed a proprietary 15-factor technical analysis scoring system integrated with a personalized trading profile engine, dynamically adapting AI outputs and automated reporting to distinct risk horizons.
- Implemented a secure Bring Your Own Keys (BYOK) cryptographic architecture with Fernet encryption, ensuring scalable, rate-limit-free AI execution across concurrent enterprise users.

CHARLIE v2.0: Local LLM Multi-Agent System ([GitLab](#))

Python, Llama 3.1, TensorRT, DINOv2, Boston Dynamics SDK, Jetson Orin Nano, Ollama

- Architected a production-ready, modular AI assistant executing local LLM inference using 8B quantized Llama 3.1 models, integrated with the NASA JPL Visual Perception Engine for real-time scene understanding.
- Engineered a capability-based **multi-agent routing architecture** combining dynamic wake-word detection, STT, and neural TTS, optimized for highly constrained, offline deployment environments.

Bio-Oracle: Neuro-Symbolic Agentic AI ([GitHub](#))

Python, PydanticAI, Gemini, Cellpose, Docker, Robust Z-score Normalization

- Developed a **Neuro-Symbolic Agent** for high-throughput phenotypic screening, automating discovery processes on the BBBC021 dataset by combining deep learning segmentation with LLM-based reasoning.
- Architected the agentic workflow using **PydanticAI** to provide structured, verifiable reasoning over biological data, packaged in a production-ready Docker container optimized for local inference.

Enterprise Document Intelligence Platform

AWS Bedrock, SageMaker, Textract, S3, Lambda, Python, LLM APIs, Streamlit

- Built scalable document processing system automating extraction from 2,500+ PDFs using LLM-powered OCR pipelines with AWS Bedrock and SageMaker, implementing RAG architecture for intelligent document querying.
- Designed prompt engineering strategies and model evaluation frameworks for multi-LLM orchestration, achieving high accuracy in regulatory document classification with governance-compliant audit trails.

EDUCATION

Master of Science, Data Science

Aug 2022 – May 2024

University of Maryland Baltimore County (UMBC) — GPA: 3.8/4.0

- **Relevant Coursework:** Machine Learning, Deep Learning, NLP, Computer Vision, Big Data Systems, Statistical Analysis, Predictive Modeling

CERTIFICATIONS & PUBLICATIONS

Professional Certifications

- **AWS Machine Learning Associate Certification** – [View Credential](#) – Active through Apr 2028
- **NVIDIA Deep Learning Institute** – Transformer-Based NLP Applications, GPU-Accelerated Computing

Publications & Research

- **Ranjan, R., Shroff, H., et al.** (2024). "FilletCam AI: Precision color profiling using deep learning." *Journal of Agriculture and Food Research*. DOI: [10.1016/j.jafr.2024.101461](#)
- **Trivedi, K., Shroff, H.** (2021). "Mosquito identification using ML on embedded systems." *IEEE ITU Kaleidoscope Conference*. DOI: [10.23919/ITUK53220.2021.9662116](#)